

Ertem Nusret Tas

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Education

09/2019— 06/2025 (<i>expected</i>)	Ph.D in Electrical Engineering Stanford University (SU) GPA: 4.0/4 Advisor: David Tse
09/2018— 09/2019	MEng in Electrical Engineering and Computer Science Massachusetts Institute of Technology (MIT) Advisor: Eytan Modiano
09/2015— 06/2018	B.S in Electrical Engineering and Computer Science Massachusetts Institute of Technology (MIT) GPA: 5.0/5
09/2015— 06/2018	Minor in Economics Massachusetts Institute of Technology (MIT)

Research

09/2019— <i>present</i>	Stanford University (SU) Security of Blockchain Systems [1, 3-19, 21, 22, 24, 25] Supervised by Prof. David Tse
09/2018— 08/2019	Massachusetts Institute of Technology (MIT) Coflow Scheduling in Cross-bar Switches [23] Supervised by Prof. Eytan Modiano
09/2016— 05/2017	Massachusetts Institute of Technology (MIT) Infrared Sensors based on Graphene Flake Films Supervised by Prof. Tomas Palacios

Experience

06/2023— 08/2023	a16z crypto research Intern	New York, NY
06/2022— 09/2022	BabylonChain Intern	Stanford, CA
06/2021— 09/2021	Celestia (Lazyledger) Intern	Stanford, CA
06/2018— 08/2018	Apple Inc. Intern at the Silicon Eng. Group Formal Verification Team	Austin, TX

06/2017—
08/2017 **NetApp Inc.** Research Triangle Park, NC
Intern

05/2016—
08/2016 **Linear Technology** North Chelmsford, MA
Intern

Teaching

10/2022—
10/2023 **Lecturer**
Stanford

- Gave three lectures on consensus protocols in Cryptocurrencies and Blockchain Technologies (CS 251) in Fall 2022 and 2023.

04/2023—
06/2023 **Teaching Assistant for Information Theory (EE 276)**
Stanford

02/2018—
05/2018 **Teaching Assistant for Math for Computer Science (6.046J)**
MIT

Publications

Preprints:

- [1] J. Neu[†], E. N. Tas[†], and D. Tse[†]. “Snap-and-Chat Protocols: Systems Aspects”. arXiv:2010.10447. Sept. 2020.
- [2] E. N. Tas, J. Adler, M. Al-Bassam, I. Khoffi, D. Tse, and N. Vaziri. “Accountable Safety for Rollups”. Presented at: *The Science of Blockchain Conference 2022 (SBC ’22)* arXiv:2210.15017. Oct. 2022.
- [3] E. N. Tas, R. Han, D. Tse, F. Yu, and K. Nazirkhanova. “Interchain Timestamping for Mesh Security”. arXiv:2305.07830. May 2023. (Online version of [20])
- [4] R. Sarenche[‡], E. N. Tas[‡], B. Monnot, C. Schwarz-Schilling, and B. Preneel. “Breaking the Balance of Power: Commitment Attacks on Ethereum’s Reward Mechanism”. arXiv:2407.19479. July 2024.
- [5] X. Dong[‡], O. S. Thyfronitis Litos[‡], E. N. Tas[‡], D. Tse[‡], R. L. Woll[‡], L. Yang[‡], and M. Yu[‡]. “Remote Staking with Economic Safety”. arXiv:2408.01896. Aug. 2024.
- [6] Y. Efron[†], E. N. Tas[†]. “Dynamically Available Common Subset”. ePrint:2025/016. Jan. 2025.

Conference Papers:

- [7] A. Dembo[‡], S. Kannan[‡], E. N. Tas[‡], D. Tse[‡], P. Viswanath[‡], X. Wang[‡], and O. Zeitouni[‡]. “Everything is a Race and Nakamoto Always Wins”. In: *Proceedings of the 2020 ACM SIGSAC Conference on Computer and Communications Security (CCS ’20)*. arXiv:2005.10484. Oct. 2020.
- [8] J. Neu[†], E. N. Tas[†], and D. Tse[†]. “Ebb-and-Flow Protocols: A Resolution of the Availability-Finality Dilemma”. In: *Proceedings of the 2021 IEEE Symposium on Security and Privacy (IEEE S&P ’21)*. arXiv:2009.04987. May. 2021.

[‡] Listed alphabetically.

[†] Contributed equally and listed alphabetically.

- [9] J. Neu[†], E. N. Tas[†], and D. Tse[†]. “The Availability-Accountability Dilemma and its Resolution via Accountability Gadgets”. In: *Proceedings of 26th International Conference on Financial Cryptography and Data Security (FC’22)*. arXiv:2105.06075. May. 2021.
- [10] C. Schwarz-Schilling, J. Neu, B. Monnot, A. Asgaonkar, E. N. Tas, and D. Tse. “Three Attacks on Proof-of-Stake Ethereum”. In: *Proceedings of 26th International Conference on Financial Cryptography and Data Security (FC’22)*. arXiv:2110.10086. Oct. 2021.
- [11] J. Neu[†] and E. N. Tas[†]. “Two Attacks On Proof-of-Stake GHOST/Ethereum”. In: *2022 Workshop on Developments in Consensus*. arXiv:2203.01315. March 2022.
- [12] E. N. Tas, D. Tse, F. Gai, S. Kannan, M.A. Maddah-Ali, and F. Yu. “Bitcoin-Enhanced Proof-of-Stake Security: Possibilities and Impossibilities”. In: *Proceedings of the 2023 IEEE Symposium on Security and Privacy (IEEE S&P ’23)*. arXiv:2207.08392. Aug. 2022.
- [13] E. N. Tas, and D. Boneh. “Cryptoeconomic Security for Data Availability Committees”. In: *Proceedings of 27th International Conference on Financial Cryptography and Data Security (FC’ 23)*. arXiv:2208.02999. Aug. 2022.
- [14] E. N. Tas, R. Han, D. Tse, and F. Yu. “Interchain Timestamping for Mesh Security”. In: *2023 ACM Conference on Computer and Communications Security (CCS ’23)*. May 2023.
- [15] S. Agrawal, J. Neu, E. N. Tas, and D. Zindros. “Proofs of Proof-of-Stake with Sublinear Complexity”. In: *Advances in Financial Technologies 2023 (AFT ’23)*. arXiv:2209.08673. Sept. 2022.
- [16] E. N. Tas, and D. Boneh. “Vector Commitments with Efficient Updates”. In: *Advances in Financial Technologies 2023 (AFT ’23)*. arXiv:2307.04085. July 2023.
- [17] E. N. Tas, D. Zindros, L. Yang, and D. Tse. “Light Clients for Lazy Blockchains”. In: *28th International Conference on Financial Cryptography and Data Security (FC’24)*. arXiv:2203.15968. March 2022.
- [18] F. D’Amato[‡], J. Neu[‡], E. N. Tas[‡], and D. Tse[‡]. “No More Attacks on Proof-of-Stake Ethereum?”. In: *28th International Conference on Financial Cryptography and Data Security (FC’24)*. arXiv:2209.03255. Sept. 2022.
- [19] J. Neu[‡], E. N. Tas[‡], and D. Tse. ”Short Paper: Accountable Safety Implies Finality”. “<https://eprint.iacr.org/2023/1301>”. In: *28th International Conference on Financial Cryptography and Data Security (FC’24)*. ePrint:2023/1301. Sept. 2023.
- [20] E. N. Tas, I. A. Seres, Y. Zhang, M. Melczer, M. Kelkar, J. Bonneau and V. Nikolaenko. “Atomic and Fair Data Exchange via Blockchain”. In: *2024 ACM SIGSAC Conference on Computer and Communications Security (CCS ’24)*. ePrint:2024/418. March 2024. (**Distinguished paper award at CCS ’24**)
- [21] E. N. Tas[‡], D. Tse[‡], and Y. Wang[‡]. “A Circuit Approach to Constructing Blockchains on Blockchains”. In: *Advances in Financial Technologies 2024 (AFT ’24)*. arXiv:2402.00220. Aug. 2024.
- [22] S. Sridhar[†], E. N. Tas[†], J. Neu, D. Zindros and D. Tse. ”Consensus Under Adversary Majority Done Right”. “<https://eprint.iacr.org/2024/1799>”. In: *29th International Conference on Financial Cryptography and Data Security (FC’25)*. ePrint:2023/1301. Nov. 2024.

MEng Thesis:

- [23] E. N. Tas. “Coflow scheduling in data center networks”. M. Eng. Thesis. Sept. 2019.

Blog Posts:

- [24] J. Neu, E. N. Tas, and D. Tse. “Avalanche Attack on Proof-of-Stake GHOST”. Ethereum Research. Jan. 2022
- [25] J. Neu, E. N. Tas, and D. Tse. “Balancing Attack: LMD Edition”. Ethereum Research. Jan. 2022
- [26] E. N. Tas, and N. Vaziri. “Rollups on Celestia”. Celestia Forum. Sep. 2021
- [27] E. N. Tas, and N. Vaziri. “Spamming Attacks on Rollups”. Celestia Forum. Sep. 2021
- [28] E. N. Tas, and N. Vaziri. “Woods Attack on Celestia”. Celestia Forum. Sep. 2021
- [29] M. Al-Bassam, E. N. Tas, and N. Vaziri. “Rollups as Sovereign Chains”. Celestia Blog. July 2022
- [30] E. N. Tas, K. Nazirkhanova, and D. Tse. “Babylon: Bitcoin Security for Cosmos and beyond”. Substack BabylonChain. Sept. 2022
- [31] E. N. Tas. “Why is Stake Unbonding so Slow?”. Substack BabylonChain. Sept. 2022
- [32] E. N. Tas, and K. Nazirkhanova. “Babylon for Fast Stake Unbonding”. Substack BabylonChain. Sept. 2022

Honors and Awards

- Silver medalist in 2014 International Physics Olympiad (IPhO).
- Ranked 1st among 2 million participants in the University Entrance Exam of Turkey.
- Landsman Undergraduate Research and Innovation Scholar at MIT SuperUROP (Undergraduate Research Opportunities Program), 2016-2017.
- Member of Mass Beta chapter of Tau Beta Pi (TBP).
- Member of Beta Theta chapter of Eta Kappa Nu. (HKN).
- Winner of the Niantic Augmented Reality Prize at 2018 HackMIT.
- Finalist for the 2022 Meta PhD Research Fellowship in Blockchain and Cryptoeconomics.

Leadership and Activities

08/2024— 08/2024	Science and Engineering of Consensus 2024 Workshop Organizer	New York, NY
06/2022— 06/2024	Information Systems Lab. (ISL) Colloquium Colloquium Organizer	Stanford, CA

<i>09/2016— 05/2017</i>	Gordon MIT Leadership Program Gordon Engineering Leader	Cambridge, MA
<i>09/2016— 05/2017</i>	MIT Turkish Student Association President	Cambridge, MA
<i>03/2018— 03/2019</i>	MIT Tau Beta Pi (TBP) Secretary and Community Service Chair	Cambridge, MA
<i>05/2018— 05/2019</i>	MIT Eta Kappa Nu (HKN) Secretary	Cambridge, MA